

2. LITERATURE REVIEW

2.1. Introduction

The literature that supports disturbance-aware zero-sum pursuit–evasion control is distributed across several related areas rather than contained within one single field. The mathematical foundation begins with differential games and linear-quadratic dynamic games, while the computational foundation is divided between model-based Riccati methods and model-free quadratic Q-learning. The disturbance-aware component then draws from persistent-input estimation, unknown-input reconstruction, offset-free control, and affine compensation. Finally, pursuit–evasion research provides the adversarial interpretation that connects the minimizing and maximizing players to a pursuer and an evader. As a result, the literature review must establish not only that each area is active, but also how these areas connect into one coherent thesis problem.

The current direction differs from a broad comparison between classical control and deep reinforcement learning. More specifically, the four algorithms considered in this thesis are structurally related. Model-based policy iteration and model-based value iteration use the known system matrices to solve the nominal zero-sum game. Model-free Q-learning policy iteration and model-free Q-learning value iteration seek the corresponding game policies from sampled transitions without using those matrices directly in the learning update. The methods therefore differ primarily in their information structure and computational procedure, not in the game being solved. Rizvi and Lin provide the immediate algorithmic organization for this four-method structure, while earlier zero-sum Q-learning work establishes the model-free connection between the quadratic action-value function and the game algebraic Riccati equation (GARE) [1], [2].

A second distinction is that the thesis does not treat the nominal algorithms as complete representations of implementation behavior. Most derivations assume that the input computed by a policy is the input that enters the system. This assumption allows the Riccati and Bellman equations to be written cleanly, but it excludes persistent command-channel mismatch. A constant acceleration bias, steering-center shift, actuator calibration error, or repeated difference between a commanded and realized input introduces an affine term into the closed-loop dynamics. The

nominal feedback gains may remain stabilizing while the realized trajectory no longer converges to the nominal equilibrium. This observation motivates the inclusion of literature on disturbance models, simultaneous input and state estimation, and offset-free control [3], [4], [5], [6].

The literature is therefore organized around six connected questions. First, what conditions define a solvable two-player zero-sum linear-quadratic game? Second, how do model-based policy iteration and value iteration compute the associated saddle solution? Third, how do quadratic Q-learning methods recover equivalent policies from sampled data? Fourth, how can persistent command offsets be represented and estimated without changing the nominal game definition? Fifth, how does pursuit–evasion provide a meaningful interpretation for the minimizing and maximizing players? Sixth, which numerical checks are necessary to distinguish a valid saddle solution from an iteration that merely stops changing?

This organization intentionally gives the greatest attention to the literature that directly supports the four algorithms and the disturbance-compensation extension. Autonomous-ground-vehicle and QCar sources remain relevant as an application pathway, but they are no longer the center of the theoretical review. The present contribution is first established on controlled discrete-time zero-sum systems so that the original algorithms, the added bias model, and the resulting compensation behavior remain mathematically traceable. Nonlinear vehicle dynamics, acceleration and steering bias vectors, QLab simulation, and physical QCar testing are retained as later extensions rather than being presented as completed validation.

2.2. Zero-Sum Linear-Quadratic Dynamic Games and the GARE

Differential game theory formalizes decision problems in which several players influence the same dynamical system while pursuing different objectives. Isaacs established the classical pursuit–evasion and differential-game framework, including the interpretation of opposing players and terminal capture conditions [7]. Basar and Olsder later developed the general theory of dynamic noncooperative games, including zero-sum, Nash, open-loop, and feedback formulations [8]. These sources remain foundational because the language used throughout this thesis—player, policy, value, saddle point, and zero-sum objective—originates from this game-theoretic structure.

The zero-sum case is distinguished by a single performance index that one player attempts

to minimize and the other attempts to maximize. For a discrete-time linear system, the nominal dynamics can be represented as

$$x_{k+1} = Ax_k + B_P u_{P,k} + B_E u_{E,k}, \quad (1)$$

where x_k is the game state, $u_{P,k}$ is the minimizing-player input, and $u_{E,k}$ is the maximizing-player input. A representative infinite-horizon quadratic cost is

$$J = \sum_{k=0}^{\infty} \left(x_k^\top Q x_k + u_{P,k}^\top R_P u_{P,k} - u_{E,k}^\top R_E u_{E,k} \right), \quad (2)$$

where $Q \succeq 0$, $R_P \succ 0$, and $R_E \succ 0$. The sign difference between the two input penalties reflects the minimax structure. The pursuer attempts to reduce the state and its own control effort, while the evader acts to increase the same game value within the admissible saddle conditions.

The linear-quadratic structure is important because it produces policies that are linear in the state and a value function that is quadratic. More specifically, if a stabilizing saddle solution exists, the value may be written as

$$V(x_k) = x_k^\top P x_k, \quad (3)$$

and the policies take the form

$$u_{P,k} = L x_k, \quad u_{E,k} = K x_k. \quad (4)$$

The resulting closed-loop matrix is $A + B_P L + B_E K$. Pachter and Pham give a discrete-time treatment of linear-quadratic dynamic games that is especially relevant because the MATLAB algorithms in this thesis operate in sampled time rather than through continuous-time differential equations [9]. The discrete-time formulation also makes the relationship between the Bellman equation, the GARE, and quadratic Q-learning direct.

The GARE is the central matrix equation for the nominal model-based problem. Its block structure contains the interactions among the state, minimizing input, and maximizing input. Unlike the single-player LQR problem, the zero-sum game must satisfy both a minimizing curvature

condition and a maximizing curvature condition. The minimizing-player control block must be positive definite, while the appropriate maximizing-player Schur complement must be negative definite. These conditions are not secondary numerical conveniences. They determine whether the stationary point obtained from the joint quadratic form is a saddle point rather than a minimum in both inputs or an indefinite stationary point.

The pursuit–evasion specialization of linear-quadratic games has a long history. Ho established an early analytical result for LQ pursuit–evasion differential games, while Bagchi and Olsder extended the setting to stochastic disturbances [10], [11]. Shinar examined solvability conditions for LQ differential games associated with pursuit–evasion problems, demonstrating that an apparently reasonable quadratic formulation does not automatically guarantee the existence of an admissible saddle solution [12]. These results support the decision in this thesis to report curvature and stability checks explicitly rather than infer validity from the existence of a numerical matrix alone.

More recent work continues to study the distinction between open-loop and feedback equilibria and the behavior of zero-sum LQ games under stochastic or long-horizon conditions. Sun and Yong distinguish open-loop and closed-loop saddle points in stochastic LQ differential games, while Sun later develops the two-person stochastic zero-sum case in greater detail [13], [14]. Monti et al. examine feedback and open-loop Nash equilibria for infinite-horizon discrete-time dynamic games, which reinforces the importance of stating which equilibrium concept is implemented by an algorithm [15]. Sun and Yong’s recent long-time analysis further demonstrates that the LQ game remains an active theoretical setting rather than only a historical benchmark [16].

The current learning literature also treats zero-sum LQ games as a demanding benchmark. Wu et al. emphasize that the corresponding optimization problem is nonconvex and nonconcave and develop improved sample-complexity and last-iterate convergence results [17]. This work is important for the present thesis because it places the model-free algorithms within the current state of research while also clarifying that convergence claims depend on the precise algorithm, information assumptions, and admissible policy region. The current thesis does not attempt to reproduce that sample-complexity theory. Instead, it uses the established LQ game structure to compare the four algorithms under a shared persistent command-offset model.

The GARE literature therefore establishes three requirements that carry into the remainder of

the thesis. First, the nominal problem must be stated as a genuine minimax game rather than two independent controllers. Second, the feedback gains must be associated with a stabilizing value matrix and valid saddle curvature. Third, any disturbance-aware extension should preserve the nominal GARE structure if the thesis claims to extend rather than replace the original algorithms. These requirements provide the basis for the model-based policy-iteration and value-iteration literature reviewed next.

2.3. Model-Based Policy Iteration and Value Iteration

Policy iteration and value iteration are related dynamic-programming procedures, but they should not be treated as interchangeable names for the same computation. Policy iteration alternates between evaluating fixed policies and improving those policies. Value iteration repeatedly updates the value function through a Bellman or Riccati mapping and then obtains the policies from the current value estimate. The two procedures may converge to the same nominal solution under compatible assumptions, but their intermediate matrices, initialization requirements, and numerical behavior can differ.

The policy-iteration lineage is connected to iterative Riccati methods developed for optimal regulation. Kleinman introduced an iterative Riccati technique based on repeated Lyapunov equations, and Hewer developed a discrete-time steady-state gain iteration [18], [19]. Although these works concern single-player regulation, they establish the basic computational principle that a nonlinear Riccati equation can be approached through a sequence of linear matrix equations associated with current policies. Mageirou extended iterative ideas to Riccati game equations, making the connection to dynamic games explicit [20].

For a zero-sum game, policy evaluation is performed with the current minimizing and maximizing gains fixed. If L_i and K_i are the policies at iteration i , then the current closed-loop matrix is

$$A_i = A + B_P L_i + B_E K_i. \quad (5)$$

Provided that A_i is Schur stable, the associated value matrix can be obtained from a discrete

Lyapunov equation. Policy improvement then updates both gains from the blocks of the game quadratic form. The two improvements are coupled because the pursuer gain depends on the evader blocks and the evader gain depends on the pursuer blocks. This coupling is one reason that a zero-sum policy iteration cannot be reduced to two independent LQR updates.

Abu-Khalaf, Lewis, and Huang developed policy iterations for the Hamilton–Jacobi–Isaacs equation in H_∞ state-feedback control with input saturation, demonstrating how the game interpretation connects disturbance attenuation and minimax control [21]. Wu and Luo later proposed simultaneous policy updates for learning continuous-time H_∞ control solutions [22]. These works are relevant because the pursuer-versus-evader interpretation and the controller-versus-disturbance interpretation share the same zero-sum mathematical structure, even when the physical meanings of the two players differ.

Rizvi and Lin organize model-based policy iteration and value iteration alongside their model-free counterparts for linear systems [1]. This organization is the direct starting point for the algorithmic structure in the current thesis. The model-based policy-iteration code uses known matrices to evaluate a current closed-loop policy and then improve the two gains. The model-based value-iteration code repeatedly applies the GARE mapping and then computes the policies associated with the updated value matrix. The present disturbance-aware work retains these roles. The command-offset estimation and compensation procedures are added to the execution and evaluation layers rather than inserted into the nominal Riccati equations without justification.

Recent research has continued to refine zero-sum policy iteration. Zhao et al. propose a single-loop policy-iteration method for linear zero-sum games and establish convergence for the resulting GARE sequence [23]. The significance of this work is not that every earlier policy iteration is obsolete. Rather, it shows that initialization, loop structure, and convergence proof remain active concerns. Li et al. similarly develop a relaxed policy-iteration algorithm for nonlinear zero-sum games with application to H_∞ control [24]. Nortmann et al. examine iterative and data-driven algorithms for linear-quadratic discrete-time dynamic games, further demonstrating the current interest in connecting model-based game solutions to data-driven computation [25].

Value iteration follows a different computational logic. Starting from an initial value matrix, the algorithm repeatedly applies a Riccati or Bellman operator. The policies are functions of the

current value matrix rather than fixed during a separate evaluation step. Bertsekas provides the broader dynamic-programming foundation for this distinction [26]. In zero-sum games, the value update must preserve the minimax block structure and remain within a region where the necessary matrix inverses or linear solves are well defined.

Guo, Wang, and Zhang develop on-policy and off-policy value-iteration algorithms for stochastic zero-sum dynamic games with unknown dynamics [27]. Their work is especially relevant because it treats both model-based convergence and model-free forms within one value-iteration setting. It also shows that the line between model-based and model-free computation is determined by the information used in the update, not simply by whether the word “value iteration” appears in the algorithm title.

The literature suggests several principles for the current thesis. First, policy iteration and value iteration should be implemented and reported separately. Second, changes made for numerical robustness, such as replacing explicit matrix inversion with a linear solve, should be identified as implementation choices rather than new control laws. Third, the original update equations should remain recognizable when the disturbance-aware layer is added. Fourth, convergence histories should report changes in the value matrix and both gains because a small value update does not automatically prove that all policy quantities have converged.

These principles also clarify the role of the model-based algorithms in the final comparison. They are not used merely to generate a single reference matrix. They provide two established procedures with different iteration structures but the same nominal system knowledge. The disturbance-aware study then asks whether the same command-offset model and compensation logic can be applied without altering what each nominal algorithm does.

2.4. Model-Free Quadratic Q-Learning and Adaptive Dynamic Programming

Model-free control is often discussed as if it refers only to deep neural-network policies trained through long episodes. That framing is too broad for the present thesis. The model-free methods considered here are structured quadratic Q-learning algorithms for linear zero-sum systems. Their purpose is to recover the value and policy information needed by the game without inserting the analytical state-transition matrices into the learning update.

Watkins and Dayan established Q-learning as a model-free temporal-difference method for estimating optimal action values [28]. Sutton and Barto provide the broader reinforcement-learning framework, including value functions, Bellman equations, exploration, and temporal-difference learning [29]. For linear-quadratic control, however, the action-value function has additional structure. Bradtke and Barto developed linear least-squares temporal-difference methods, and Lagoudakis and Parr later developed least-squares policy iteration [30], [31]. These sources establish the general connection between sampled transitions, linear parameter estimation, and policy improvement.

In a two-player zero-sum linear-quadratic game, the joint state-action vector can be written as

$$\zeta_k = \begin{bmatrix} x_k^\top & u_{P,k}^\top & u_{E,k}^\top \end{bmatrix}^\top, \quad (6)$$

and the quadratic action-value function is represented by

$$Q_H(\zeta_k) = \zeta_k^\top H \zeta_k, \quad (7)$$

where H is symmetric. The state, pursuer-input, evader-input, and cross-coupling blocks of H contain the information needed to recover the two saddle policies. The learner therefore estimates a finite set of quadratic coefficients rather than an unrestricted nonlinear function.

Al-Tamimi, Lewis, and Abu-Khalaf provide foundational adaptive-critic and model-free Q-learning designs for discrete-time zero-sum games with application to H_∞ control [2], [32]. Their work is central because it demonstrates that the GARE solution can be recovered from data through a quadratic Q-function without direct knowledge of the system matrices. Lewis and Vrabie place these ideas within adaptive dynamic programming and feedback control more generally [33]. These sources form the conceptual bridge from nominal Riccati computation to the model-free algorithms in this thesis.

Policy-iteration Q-learning evaluates the Q-function associated with current pursuer and evader policies and then improves the policies from the estimated matrix partitions. Luo et al. develop a policy-iteration Q-learning method for data-based two-player zero-sum linear discrete-time systems [34]. The method emphasizes that the data matrix must contain sufficient independent

quadratic information. In practical terms, the samples must excite the state and both input channels strongly enough that the symmetric parameters of H can be identified.

Finite-horizon and output-feedback extensions broaden this structured learning framework. Liu et al. develop finite-horizon Q-learning for discrete-time zero-sum games and later output-feedback Q-learning for finite-horizon problems [35], [36]. These studies are relevant because they show that the quadratic zero-sum learning structure is not limited to one infinite-horizon full-state formulation. Lian et al. consider off-policy inverse Q-learning for unknown antagonistic systems, while Guo et al. develop Q-learning for stochastic zero-sum games [37], [38].

Recent research extends the same ideas toward nonlinear, switched, delayed, and safety-constrained systems. Wang et al. propose neural Q-learning for discrete-time nonlinear zero-sum games with an adjustable convergence rate [39]. Wang et al. also develop a parallel off-policy Q-learning method for discrete-time Markov jump systems [40]. Jiang et al. address delayed two-player zero-sum games through data-based Q-learning, and Liu et al. study safe reinforcement learning for nonlinear zero-sum games with unknown state constraints and asymmetric input constraints [41], [42]. These papers represent current developments, but they also help define the boundary of the present work. The thesis does not introduce neural approximators, Markov jumps, unknown safety sets, or delayed dynamics into the four baseline algorithms. Instead, it develops a controlled persistent-offset extension whose effect can be isolated.

The literature also distinguishes structured zero-sum Q-learning from policy-gradient optimization. Zhang, Yang, and Basar establish convergence results for policy optimization in zero-sum LQ games, while Wu et al. later improve sample-complexity and last-iterate guarantees [17], [43]. These works are important state-of-the-art references, but they do not replace the quadratic Q-learning algorithms used as the thesis starting point. They show that zero-sum LQ learning can be approached through different computational routes, each with different theoretical assumptions.

Several numerical requirements follow directly from the model-free literature. The feature matrix must have sufficient rank to identify the symmetric Q-function parameters. The regression matrix must remain adequately conditioned. The behavior policies must provide excitation without driving the system outside the region in which the data are meaningful. The estimated game matrix must satisfy the same minimizing and maximizing curvature requirements used by the model-based

solution. Finally, a small Bellman residual should be interpreted together with gain error, rank, conditioning, and closed-loop behavior rather than used as a stand-alone proof of policy recovery.

The present thesis uses these principles to keep the model-free comparison disciplined. The model-free label refers to the absence of the analytical transition matrices from the Q-learning update. It does not imply that no measurements, no prior structural assumptions, or no commanded-versus-realized input data are available. This distinction becomes especially important when persistent command offsets are introduced, because the policy may be learned without the matrices while the bias estimate may still use measured actuator-channel information.

2.5. Persistent Command Bias, Unknown-Input Estimation, and Affine Compensation

The disturbance-aware contribution begins with a simple but consequential observation: a control command and a realized control input are not always identical. A persistent mismatch may be caused by actuator calibration, steering-center error, longitudinal response offset, conversion scaling, or another repeatable implementation effect. If the mismatch remains approximately constant over a calibration batch and the following run, it can be represented as an additive command-channel bias.

For the two-player game, the realized inputs are modeled as

$$u_{P,k}^{\text{act}} = u_{P,k}^{\text{cmd}} + b_P, \quad u_{E,k}^{\text{act}} = u_{E,k}^{\text{cmd}} + b_E. \quad (8)$$

When the nominal policies are applied without compensation, the closed-loop dynamics become

$$x_{k+1} = (A + B_P L + B_E K)x_k + B_P b_P + B_E b_E. \quad (9)$$

The additional term is affine and persistent. If the nominal closed-loop matrix is Schur stable, the state need not diverge; instead, it may approach a nonzero equilibrium determined by the bias and the closed-loop resolvent. This behavior is important because a nominal stability result can coexist with a meaningful trajectory error.

The offset-free control literature provides a mature foundation for representing this problem.

Muske and Badgwell examine disturbance modeling for offset-free linear model predictive control, while Pannocchia and Rawlings develop disturbance models that remove steady-state tracking offset [3], [4]. Maeder et al. provide a systematic treatment of linear offset-free MPC, and Jimoh et al. compare different offset-free formulations [44], [45]. Although these studies are primarily written in the MPC setting, the underlying principle is directly relevant: persistent mismatch should be represented explicitly rather than treated as random numerical noise.

The unknown-input estimation literature addresses the related problem of reconstructing unmeasured forcing terms. Gillijns and De Moor derive unbiased minimum-variance input and state estimators for linear discrete-time systems [5]. Yong et al. later provide a unified filter for simultaneous input and state estimation [6]. These methods are more general than the batch constant-bias estimator used in the present model-based MATLAB benchmark. Nevertheless, they establish the identifiability and estimation principles that justify treating a persistent command offset as an estimable input rather than an arbitrary unexplained deviation.

For the known-model case, the transition residual is

$$r_k = x_{k+1} - Ax_k - B_P u_{P,k}^{\text{cmd}} - B_E u_{E,k}^{\text{cmd}}. \quad (10)$$

Under the bias model, this residual satisfies

$$r_k = \begin{bmatrix} B_P & B_E \end{bmatrix} \begin{bmatrix} b_P \\ b_E \end{bmatrix} + \eta_k, \quad (11)$$

where η_k represents measurement noise, process noise, and finite-precision effects. A batch least-squares estimate is possible when the combined bias map has sufficient rank. This requirement is physically meaningful. If the two input channels affect the state in indistinguishable directions, the individual player biases cannot be separated from state-transition data alone.

Once the offsets are estimated, affine command compensation subtracts the estimates from the nominal policy outputs:

$$u_{P,k}^{\text{cmd}} = Lx_k - \hat{b}_P, \quad u_{E,k}^{\text{cmd}} = Kx_k - \hat{b}_E. \quad (12)$$

The realized dynamics then become

$$x_{k+1} = (A + B_P L + B_E K)x_k + B_P(b_P - \hat{b}_P) + B_E(b_E - \hat{b}_E). \quad (13)$$

Exact estimates recover the nominal closed-loop dynamics, while imperfect estimates leave only the residual bias. This relationship provides a direct interpretation of the numerical results. Compensation performance should be consistent with the remaining estimation error rather than described as an unexplained improvement.

Recent work continues to study identification and offset-free stability under more general conditions. Berberich et al. develop data-driven tracking MPC for nonlinear systems, showing how steady-state tracking and model-free data can be connected [46]. Kuntz and Rawlings examine maximum-likelihood identification with integrating disturbances and offset-free stability under plant-model mismatch [47], [48]. Yang et al. consider joint state and bias estimation for nonlinear systems under non-Gaussian process noise [49]. These studies extend beyond the constant-bias benchmark, but they show that bias estimation and offset-free control remain current research topics.

The literature also clarifies what is and is not claimed by the present compensation method. The estimator is not a universal unknown-input observer, and it does not guarantee identification under arbitrary time-varying disturbances. The compensation law does not redesign the saddle policy or solve a new robust game. It assumes that the nominal policy has already been obtained and that the persistent command mismatch can be estimated over a calibration interval. The resulting affine correction is therefore best described as a disturbance-aware execution layer attached to an established nominal algorithm.

This distinction is central to the thesis contribution. The policy-iteration, value-iteration, and Q-learning equations retain their original mathematical roles. The added work defines how the commanded policies are translated into realized inputs, how persistent offsets are estimated under the information available to each method, and how compensation quality is measured through nominal, uncompensated, estimated-compensation, and exact-compensation cases.

2.6. Pursuit–Evasion Control and Current Disturbance-Aware Learning

Pursuit–evasion provides the physical interpretation for the two-player zero-sum game. The pursuer seeks to reduce a relative-state cost or drive the engagement into a capture set, while the evader attempts to increase the same objective or delay capture. Weintraub, Pachter, and Garcia provide a modern tutorial that connects pursuit–evasion differential games to guidance, reachability, and control applications [50]. Bopardikar, Bullo, and Hespanha show how sensing limitations alter discrete-time pursuit–evasion behavior, which is relevant because the current algorithms are implemented in sampled time and later extensions may not assume complete measurements [51].

The classical pursuit–evasion literature often focuses on geometric barriers, terminal capture conditions, or specific vehicle classes. The present thesis uses pursuit–evasion in a narrower way. The minimizing player is interpreted as the pursuer and the maximizing player as the evader, but the initial MATLAB studies are designed to validate the zero-sum algorithms and the command-offset extension before introducing a full nonlinear engagement. This approach preserves the application meaning while avoiding a claim that a low-dimensional benchmark reproduces every property of a car, aircraft, spacecraft, or mobile robot.

Recent surveys show that reinforcement learning has become increasingly common in pursuit–evasion research. Yang et al. review RL approaches across pursuit–evasion applications and identify value-based, policy-gradient, multi-agent, and hybrid methods [52]. Their review is useful for locating the current thesis within the field, but the thesis does not attempt to survey or compare all modern RL architectures. The focus remains on quadratic Q-learning methods that preserve a direct connection to the zero-sum Riccati solution.

Adaptive dynamic programming has also been applied directly to pursuit–evasion problems. Gong et al. develop an ADP solution for agent pursuit–evasion, and Zhang et al. propose fixed-time zero-sum pursuit–evasion control for multisatellite systems through ADP [53], [54]. Xu et al. develop integral reinforcement-learning-based optimal pursuit–evasion control for multiple quadrotor vehicles, while Li et al. combine ADP-based pursuit–evasion control with obstacle avoidance [55], [56]. These studies demonstrate that structured game-learning methods remain active even as deep RL becomes more visible.

Current work has also moved toward constraints, safety, uncertainty, and physical realism. Kokolakis et al. study safety-aware pursuit–evasion in unknown environments using Gaussian processes and finite-time convergent reinforcement learning [57]. Gonultas and Isler investigate learning under dynamic and sensor constraints and include physical robotic evaluation [58]. Wang et al. develop a self-play iteration for aircraft pursuit–evasion games, while Chen et al. examine online multi-UAV planning in unknown environments [59], [60]. Wang et al. further emphasize safety, stability, and robustness in pursuit–evasion differential games [61].

These recent papers identify the broader frontier but also help define the present research gap. Many pursuit–evasion learning studies introduce nonlinear dynamics, obstacles, sensing limitations, self-play, or multi-agent coordination. Those additions are important, but they make it difficult to isolate whether performance differences originate from the game solution method, the approximation architecture, the exploration procedure, or the environment. The current thesis therefore takes a controlled complementary approach. It begins with a shared linear-quadratic game, compares four closely related algorithms, and introduces one structured implementation disturbance whose mathematical effect can be stated exactly.

The pursuit–evasion literature also supports the longer-term vehicle direction. A later autonomous-ground-vehicle implementation can interpret b_P and b_E as vectors containing acceleration and steering offsets. The nominal state can be replaced by relative position, heading, and speed errors derived from kinematic bicycle dynamics. Rajamani provides the vehicle-dynamics foundation for such an extension, while Tiriolo et al. demonstrate the use of a QCar platform for input-constrained autonomous-vehicle control [62], [63]. Quanser documentation then defines the practical command and measurement interfaces for future QCar experiments [64], [65]. These sources remain relevant, but they support future transfer rather than the principal theoretical claims of the present chapter.

As a result, pursuit–evasion serves three roles in the thesis. First, it gives the minimizing and maximizing players a consistent interpretation. Second, it motivates why disturbances affecting either input channel matter strategically rather than only as single-agent tracking errors. Third, it provides a clear pathway from the linear benchmark to later vehicle experiments. The literature does not require the thesis to claim completed hardware validation in order for the pursuit–evasion

interpretation to be meaningful.

2.7. Numerical Well-Posedness and Reproducible MATLAB Validation

The literature review would be incomplete if it treated numerical implementation as a separate programming issue. The four algorithms depend on matrix solves, Lyapunov equations, quadratic regressions, block Hessians, and repeated gain updates. A mathematically valid equation may still produce misleading numerical results if the relevant matrices are singular, poorly conditioned, or evaluated outside the admissible policy region.

For the model-based methods, three checks are especially important. First, the current closed-loop matrix used during policy evaluation must be Schur stable so that the discrete Lyapunov equation has the intended stabilizing solution. Second, the final GARE residual must be small relative to the scale of the value matrix. Third, the game curvature conditions must hold. The pursuer block must be positive definite, and the evader Schur complement must be negative definite. A low iteration difference without these checks does not establish a valid saddle solution.

For the model-free methods, the numerical requirements are broader. The quadratic feature matrix must have sufficient column rank. The least-squares problem must be adequately conditioned. The estimated H matrix should be symmetrized only to remove roundoff-level asymmetry rather than to hide a structurally inconsistent regression. The recovered policies must satisfy the game curvature conditions, and the resulting closed-loop dynamics must be checked. Horn and Johnson provide the matrix-analysis foundation for eigenvalue, definiteness, Schur-complement, and conditioning arguments [66].

The use of reciprocal condition estimates and linear solves is therefore part of the numerical verification. Replacing an explicit inverse by MATLAB's backslash operator does not change the underlying equation; it changes how the equation is evaluated. This distinction is important when documenting extensions to existing code. A numerically safer solve should not be described as a new algorithm, while a change to the order or meaning of the policy updates would be algorithmic.

The disturbance-estimation problem adds its own rank and conditioning requirements. The combined bias matrix $[B_P \ B_E]$ must distinguish the two command channels. The calibration residual should be reported so that an accurate state trajectory is not attributed to a bias estimate

that was never validated. The exact-compensation case provides an additional implementation check: when the true biases are subtracted, the nominal and compensated trajectories should agree to numerical precision. If they do not, then the compensation has been inserted inconsistently with the realized input model.

Reproducibility further requires that the MATLAB scripts record more than the final gains. The relevant outputs include convergence histories, final residuals, curvature values, spectral radius, true and estimated biases, calibration error, nominal and disturbed trajectories, predicted steady-state offsets, finite-window cost, and exact-compensation error. Saving figures, summary tables, and complete workspaces allows the numerical claims to be traced back to one fixed run.

This approach is consistent with the broader movement toward reproducible benchmark studies. The value of the linear zero-sum problem is not that it is the most complicated pursuit–evasion environment. Its value is that the expected structure is known well enough to expose errors in implementation. A benchmark that reports only an attractive trajectory cannot distinguish a valid game solution from an accidental closed-loop response. By contrast, matched numerical checks allow each stage of the algorithm and compensation layer to be inspected.

2.8. Synthesis and Research Gap

The literature establishes a mature foundation for nominal zero-sum control. Differential-game theory defines the minimax problem, LQ game theory connects the saddle solution to Riccati equations, and model-based policy iteration and value iteration provide established numerical procedures. Quadratic Q-learning and adaptive dynamic programming then recover corresponding policies from sampled transitions under rank, excitation, and conditioning requirements. Pursuit–evasion gives these methods an adversarial control interpretation, while recent research extends them toward nonlinear dynamics, safety constraints, sensor limitations, and multi-agent systems.

At the same time, the reviewed literature reveals a separation among these areas. The nominal algorithm papers generally assume that commanded inputs are realized exactly. Offset-free and unknown-input estimation papers address persistent mismatch, but they are usually developed for tracking control, MPC, or state estimation rather than a matched four-algorithm zero-sum comparison. Pursuit–evasion learning papers increasingly address safety and disturbances, but they

often introduce nonlinear function approximators, self-play, obstacles, or multi-agent coordination before establishing a common model-based and model-free benchmark.

The specific gap addressed by this thesis is therefore not the absence of policy iteration, value iteration, Q-learning, or disturbance estimation individually. Each of those areas is well established. The gap lies in applying a single persistent command-channel model, estimation logic, affine compensation law, and numerical validation framework to four structurally related zero-sum algorithms while preserving the original nominal update equations.

More specifically, the literature does not commonly provide a matched study that contains all of the following elements:

- 1) a model-based GARE policy-iteration method;
- 2) a model-based GARE value-iteration method;
- 3) a model-free quadratic Q-learning policy-iteration method;
- 4) a model-free quadratic Q-learning value-iteration method;
- 5) persistent command offsets affecting both players;
- 6) bias estimation consistent with the information structure of each method;
- 7) affine compensation applied at the command channel;
- 8) nominal, uncompensated, estimated-compensation, and exact-compensation cases; and
- 9) common stability, curvature, conditioning, convergence, and trajectory diagnostics.

The thesis addresses this gap by beginning with the established algorithms documented by Rizvi and Lin and then extending their MATLAB implementation in a traceable manner [1]. The nominal Riccati, policy, and Q-learning relationships remain the algorithmic foundation. The new work defines the disturbance model, determines how the offsets are estimated, modifies the executed commands, and establishes matched numerical evidence for the resulting behavior.

This position also clarifies the role of future work. A nonlinear kinematic bicycle model, acceleration and steering bias vectors, command delay, actuator saturation, output feedback, QLABS

simulation, and physical QCar testing can be added after the four-algorithm disturbance-aware foundation is established. These extensions are meaningful because the current framework separates the nominal game solution from the implementation disturbance. The literature therefore supports a staged research program rather than requiring every possible vehicle and learning complication to be included in one thesis.

In summary, the state of the field supports both the feasibility and the need for the present study. The foundational theory explains why the four algorithms should be related. Current policy-iteration, value-iteration, and zero-sum learning research shows that the problem remains active. Disturbance-estimation and offset-free control literature provides the tools needed to represent persistent command mismatch. Recent pursuit–evasion studies demonstrate the growing importance of safety, uncertainty, and implementation constraints. The thesis connects these bodies of work through a controlled disturbance-aware zero-sum framework whose equations, assumptions, and MATLAB results remain directly traceable.

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